Data Structure and Algorithms 2 Lab

Assignment-1
Marks- 20

Problem-1: The Hedge Fund Risk Analyzer (MAX SUM / MIN SUM SUBARRAY)

Problem-Statement:

You are a quant developer at Northbridge Capital, a hedge fund that trades a single volatile asset. Every trading day the fund's position closes with either a profit or a loss. Over **N consecutive trading days**, you are given a 1D integer array D of size N, where **D[i] represents the fund's profit/loss on day i, in thousands of dollars (k$).**

- A positive integer means the fund made money that day.
- A negative integer means the fund lost money that day.

The risk committee needs two numbers before the next board meeting:

1. **Best Winning Streak:** To advertise to investors, the fund wants to know the contiguous run of trading days that produced the highest total profit.
2. **Required Capital Reserve**: To survive the worst possible losing streak without a margin call, the fund must know the size of its worst continuous run of losses (the minimum contiguous sum). The fund must hold cash equal to the absolute value of this worst streak.

Tasks Given the array **D**, calculate and output:

1. **Max_Profit_Streak**: The maximum sum of any contiguous subarray.
2. **Risk_Reserve**: The absolute value of the minimum sum of any contiguous subarray.

(Note: If the minimum subarray sum is positive: meaning the fund never had an aggregate losing streak, the required capital reserve is 0.)

Input Format

- The first line contains a single integer N, representing the number of trading days recorded.
- The second line contains N space-separated integers representing the daily_pnl array.

Output Format:

Print exactly two lines

- Line 1: An integer representing Max_Profit_Streak.
- Line 2: An integer representing Risk_Reserve.

Sample Test Cases

1. ==========

Sample Input

```
9

7 -3 12 -8 15 9 -20 4 6
```

Sample Output

```
32 20
```

Explanation:

Max_Profit_Streak: The best contiguous run is days 0 to 5: [7, -3, 12, -8, 15, 9]. The sum is 7 + (-3) + 12 + (-8) + 15 + 9 = 32.

Risk_Reserve: The worst contiguous run is the single day at index 6: [-20]. Its sum is -20, so the required reserve is 20.

2. ==========

Sample Input

```
4

-4 -10 -2 -7
```

Sample Output

```
-2 23
```

Explanation

Max_Profit_Streak: Every day is a loss, so the best the fund can do is its single least-bad day, index 2: [-2].

The maximum contiguous sum is -2. Risk_Reserve: The entire array [-4, -10, -2, -7] is a continuous drain. The minimum sum is -4 + (-10) + (-2) + (-7) = -23, so the reserve required is 23.

3. ==========

Sample Input

```
4

6 3 9 4
```

Sample Output

```
22 0
```

Explanation

Max_Profit_Streak: The entire array is profitable: [6, 3, 9, 4]. The sum is 22.

Risk_Reserve: The minimum subarray sum is 3 (a single day). Since the fund never drops into an aggregate loss, no capital reserve is required. The output is 0.

Problem-2: The Descending Merge Sort

Develop a merge sort algorithm to sort an array in descending order.

Sample Input 1

```
8
450 120 880 1000 230 880 50 900
```

Sample Output 1

```
1000 900 880 880 450 230 120 50
```